CAMPUSAI

COMPUTER NETWORKS AND INTERNET PROTOCOLS (CNIP) – SAMPLE NOTES

1. Introduction

A computer network is a group of connected devices that communicate and share data and resources. A network protocol is a set of rules that controls how devices exchange information.

2. Types of Networks

- **PAN:** Personal Area Network; covers a very short distance, such as devices connected around one person.
- **LAN:** Local Area Network; connects devices in a home, office, lab or college campus.
- **MAN:** Metropolitan Area Network; covers an area such as a city.
- **WAN:** Wide Area Network; connects networks over large geographical areas.

3. OSI Model

The OSI model explains network communication in seven layers: **Physical, Data Link, Network, Transport, Session, Presentation and Application.** Each layer performs a specific function to help data move between devices.

4. Important Internet Protocols

- **IP (Internet Protocol):** Provides addressing and routing so data can travel between networks.
- **TCP (Transmission Control Protocol):** Provides reliable, ordered delivery of data.
- **UDP (User Datagram Protocol):** Sends data with low overhead but does not guarantee delivery.
- **HTTP / HTTPS:** Used to transfer web content; HTTPS protects communication with encryption.
- **DNS:** Translates domain names, such as example.com, into IP addresses.
- **DHCP:** Automatically assigns network configuration, including IP addresses, to devices.

5. Example

When a student opens a college website, DNS helps find the server's IP address and HTTPS can securely transfer the webpage to the student's device.

6. Quick Revision Questions

1. Differentiate LAN and WAN.
2. Name the seven layers of the OSI model.
3. What is the difference between TCP and UDP?
4. What are the functions of DNS and DHCP?

Note: These are introductory sample notes. Refer to your course syllabus and class materials for complete coverage.